Road-Crash Analysis

Historical team report

Original COMP20008 Assignment 2 coursework, University of Melbourne, 2025.

Individual contribution

Kunze (Charlotte) Lyu was responsible for the clustering module and the clustering sections of the report and presentation. Data preprocessing and supervised modelling were team contributions.

Team: Tianrui Chen, Xinping Ouyang, Kunze Lyu and Na Yu.

How to read this report

This document preserves the original team submission. Historical code, CSV outputs and narrative cluster profiles were inconsistent. The cluster claims in this report have not been reproduced and should not be treated as current validated results.

The portfolio reconstruction uses 400,863 cleaned rows and reports new K=3/4/5 results in the repository README and results/recomputed/. Its findings supersede the historical cluster interpretation for the portfolio. Clusters describe feature patterns and do not establish causal injury risk.

The slides are also retained as historical presentation material. They contain older metrics and interpretations. In particular, a high silhouette score quoted historically is not the score measured in the portfolio reconstruction.

Public copy

The original author/contact block on page 1 has been removed to exclude student numbers and university email addresses. Author attribution is retained here. The remaining original report pages and analytical content are preserved. The source files were not modified.

Current evidence and review notes: https://github.com/xialuote77/road-crash-segmentation

Understanding Injury Risk on the Road: A Cluster-Based Analysis of Behavioural and Structural Factors

Group W {13}G{6}

1 Executive summary:

This project investigates how seatbelt usage, seating position, and vehicle type relate to the severity of injuries in traffic accidents. Using a merged dataset of individuals and vehicles involved in crashed, we cleaned and categorized key variables such as injury severity, seat location and vehicle body style. We applied classification methods including Logistic Regression and Random Forest to model the probability of severe injuries. In addition, clustering and mutual information analysis helped identify high-risk road users based on behavioural and vehicle-related features. Key findings suggest that not wearing a seatbelt and sitting in rear or passenger seats are associated with more serious injuries. We recommend promoting seatbelt use in all seating positions and tailoring vehicle safety messages based on user profiles.

2 Introduction:

The purpose of this report is to explore key factors that influence the severity of injuries sustained in traffic accidents. To be more specific, we focus on how seatbelt usage, seating position, and vehicle type are associated with different levels of injury severity. Understanding these factors is crucial for improving road safety, identifying vulnerable road users, and guiding policy decisions that aim to reduce the number and impact of traffic-related injuries.

This report investigates how the use of seat belts—such as whether they were worn or not—relates to injury outcomes. It also analyses how the position of the passenger within the vehicle, for example, sitting in the front versus sitting in the rear, affects injury severity. Additionally, we examine the role of various vehicle types in accidents, including common passenger cars, taxis, utilities, station wagons, panel vans, prime movers, heavy vehicles, and other light commercial vehicles. By combining these factors, we aim to identify patterns that help pinpoint high-risk road users. Clustering techniques are also applied to detect groups of individuals who share similar characteristics and face higher injury risks in crashes.

To conduct this analysis, we used four CSV datasets:

1. The **person** dataset, which contains information about individuals involved in crashes, such as injury level, seat position, and whether a seatbelt or helmet was used.
2. The **vehicle** dataset, which includes general information about the vehicles involved, such a body style and vehicle classification.

These datasets were merged using unique identifiers such as *ACCIDENT_NO* and *VEHICLE_ID*, allowing us to construct a comprehensive and structured dataset for analysis. The combined data enabled us to analyse injury severity across various combinations of user behaviour and vehicle characteristics. The analysis was motivated by the increasing need for evidence-based insights to enhance public safety and reduce the long-term social and economic costs associated with road trauma.

3 Methodology

3.1 Data Preprocessing

This study explores the influence of seat belt usage, seat position and vehicle type on the severity of injuries in traffic accidents, and analyses whether these factors can help identify high-risk road users. The analysis process includes data collection, cleaning, feature engineering, exploratory data analysis (EDA), and statistical testing.

We used two datasets, person.csv and vehicle.csv, and merged them using the shared column ACCIDENT_NO and VEHICLE_ID and filtered out the variables needed for the analysis. Including whether the seat belt is worn (HELMET_BELT_WORN), the seat position (SEATING_POSITION), the severity of the injury (INJ_LEVEL_DESC), and the vehicle type (VEHICLE_BODY_STYLE).

To ensure data integrity and reliability, missing values and outliers were dealt with. Among them, the HELMET_BELT_WORN variable uses mode imputation to reduce the influence of the missing on the result; The missing values of SEATING_POSITION and VEHICLE_BODY_STYLE are encoded as 'Unknown' to retain more samples and reduce bias.

In feature engineering, to simplify the analysis and enhance interpretability, the usage of seat belts is classified into wearing, not wearing and not applicable. The seat positions are classified as driver, front passenger, rear passenger and motorcycle passenger. The vehicle types are classified as sedans, SUVs/four-wheel drive vehicles, light commercial vehicles and motorcycles. The severity of injuries is uniformly classified into four categories: fatal injuries, serious injuries, other injuries and no injuries. Subsequently, the categorical variables were converted into numerical variables through One-Hot encoding and standardized using StandardScaler to ensure that the weights of each variable were consistent in the analysis (Pedregosa, et al., 2011).

3.2 Exploratory Data Analysis and Statistical Testing

In exploratory data analysis (EDA), the relationships among variables are visually and intuitively presented, including the correlations between seat belt usage, seat position, and vehicle type and the severity of injury. The pie chart is used to show the proportion distribution of seat belt usage, injury severity, seat distribution and vehicle type, which is convenient for understanding the data characteristics. The correlations among the variables were further quantified by calculating the standardized mutual Information (NMI) and drawing the heat map.

In this report, we used various visualisation techniques to better understand the relationships between key variables and injury severity. Among various graph types, we chose grouped bar chart, pie chart, heat map, album method and scattered bar chart. Each chart was selected based on its ability to effectively communicate patterns in the data.

Grouped bar charts are useful for comparing multiple categorical variables side by side. We used this chart to show the relationship between vehicle type, seat position, and seatbelt use, which allowed us to identify combinations (e.g., rear seat in heavy trucks without belts) that are associated with higher injury risk.

Pie charts provide a clear overview of proportions. They were applied to show the distribution of vehicle types, seat positions, and injury severity levels, giving a quick and intuitive sense of which categories dominate the dataset.

Heatmaps are excellent for highlighting correlation patterns. In this report, it was used to show the relationship between injury severity and clustering groups, vehicle type and severity category, helping us detect which clusters are more associated with severe injuries.

The elbow method was used to find the optimal number of clusters in K-means clustering. It avoids over-clustering by identifying the point where adding more clusters no longer significantly improves model fit.

Finally, scatter bar charts were used to illustrate how different clusters relate to injury severity, allowing for a visual comparison across risk groups.

These visualisations supported our analysis by making complex relationships easier to interpret and explain.

The statistical significance among the variables was verified through the Chi-Squared independence Test. The test results showed that $p = 0 < 0.05$, indicating that there was a significant statistical relationship between the seat position and the severity of injury, that is, there were obvious differences in the injury risk corresponding to different seat positions.

3.3 Clustering

In this study, we sought to identify high-risk road user profiles based on behavioural and vehicle features. To this end, we performed K-Means clustering (Pedregosa et al., 2011) using three variables central to our research question: seatbelt usage, seating position and vehicle type. Initially, the Elbow Method was applied to determine the optimal number of clusters; however, the resulting curve did not demonstrate a clear "elbow"(Figure 1), which resulted in ambiguous interpretation. In order to resolve this issue, the Silhouette Score (Rousseeuw, 1987) was computed, in which intra-cluster cohesion and inter-cluster separation are evaluated. The Silhouette Score was evaluated for $k = 3$ and $k = 4$, resulting 0.8385 and 0.1109 respectively. This finding serves to corroborate the hypothesis that the utilisation of three clusters constitutes the most interpretable and well-separated solution.

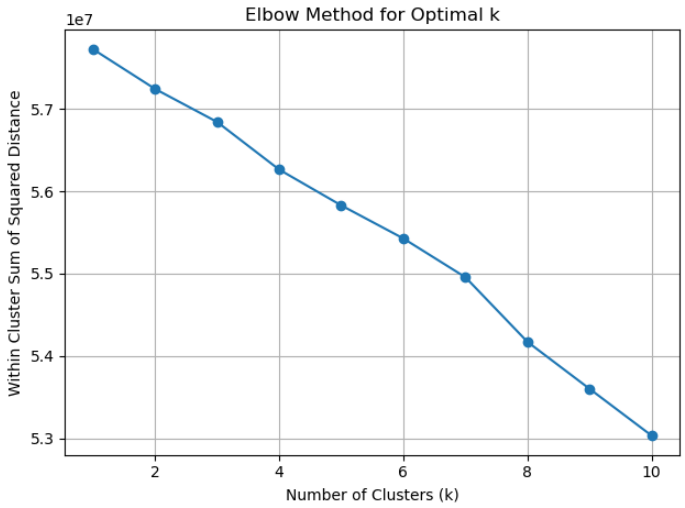


Figure 1: Elbow Method plot for KMeans clustering (1 to 10)

An investigation was then conducted into the compositions of the clusters to discover the dominant characteristics and risk levels in each group. Heatmaps and stacked bar charts were employed to visualise patterns of injury severity in relation to seatbelt usage and vehicle types across clusters. This facilitated the extraction of behavioural and vehicular types associated with elevated injury risks, thereby enabling the identification of significant subgroups within the dataset.

3.4 Supervised Learning

The dataset used for learning models is a pre-processed version of a combination of the original datasets, saved as 'cleaned_data.csv' which includes categorical columns and target column. The categorical variables were encoded using one-hot encoding which creates binary columns for each category and the target variable was converted into numerical integers using label encoding.

Two supervised learning models were selected:

- Logistic Regression
  Logistic Regression is a well-established baseline classifier that performs well when the relationship between features and the target is linear. It also provides interpretable coefficients that help assess feature importance.
- Random Forest Classifier
  Random Forest is a robust, non-linear ensemble method that can capture complex patterns in the

data. It builds multiple decision trees and combines their predictions to improve accuracy. It is less prone to overfitting than a single decision tree and provides built-in feature importance analysis.

This combination of linear and non-linear models allows us to compare performance across different learning strategies and complexity levels.

To address imbalance, class weighting was applied during training (class_weight='balanced') and stratified train-test split was applied to preserve the distribution of severity classes across both training and testing sets. In contrast to standard K-Fold cross validation, stratified splitting preserves the proportion of each class label in both sets. This is particularly important for evaluating model performance on minority classes, and it helps provide a fairer and more reliable estimate of model generalisation.

The model performance was evaluated using cross validation and different metrics such as accuracy, recall, precision and F1-score. Given the data was imbalanced, F1-score and recall were especially considered for minority class performance. The confusion matrix was also used to gain deeper insight into how the models performed across different severity classes.

4 Analysis

4.1 Exploratory Data Analysis

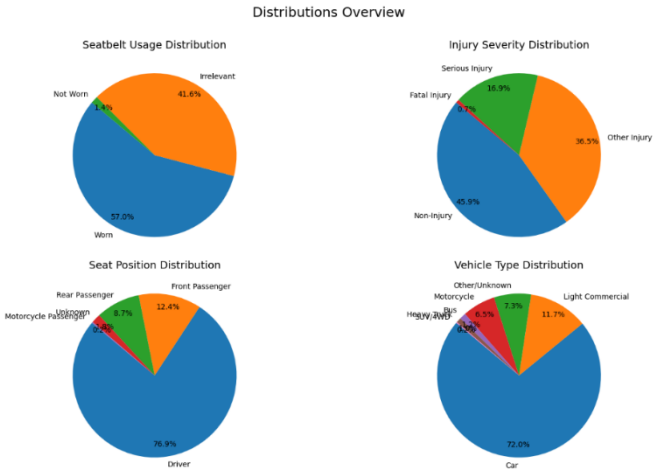


Figure 2. seatbelt Usage, Seat Position, Injury Severity, and Vehicle Type Distributions

These are pie charts showing the distributions of seatbelt usage, injury severity, seat position, and vehicle type. Each chart helps us understand the general structure of the dataset and its relevance to our research question, which investigates how seatbelt use, seat position, and vehicle type are associated with injury severity in traffic accidents.

In the Seatbelt Usage Distribution chart, we see that 57.0% of individuals wore a seatbelt, 1.4% did not, and 41.6% are classified as "irrelevant" (such as motorcycle riders, where seatbelts are not applicable). This indicates that while the majority of people comply with seatbelt usage, there is still a small group of unbelted individuals. The high proportion of "irrelevant" data highlights the importance of categorizing users by vehicle type before further analysis. This chart directly relates to our research objective, as seatbelt usage is one of the key factors we analyse in relation to injury severity.

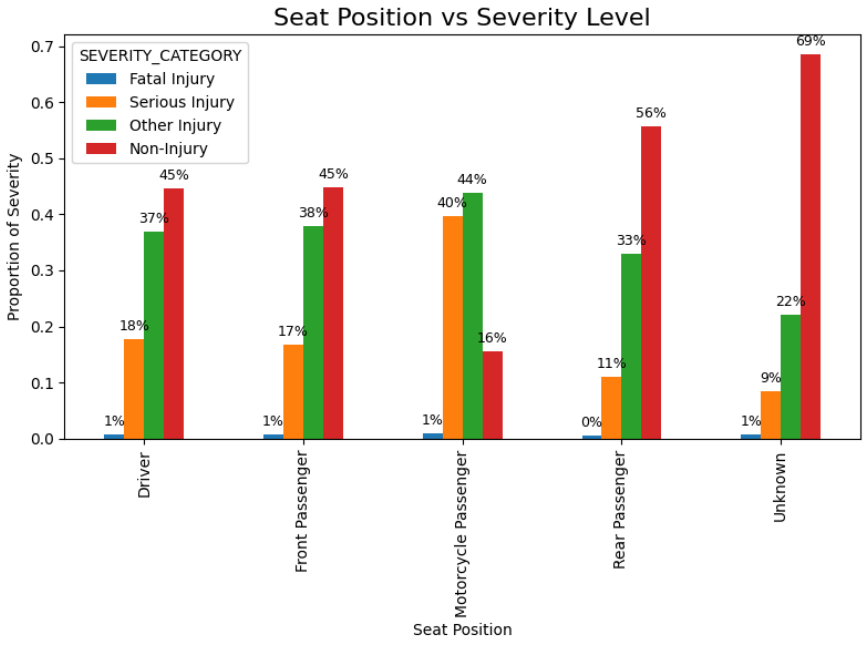
From the Injury Severity Distribution chart, we observe that 45.9% of cases resulted in no injury, 36.5% in other (minor) injuries, 16.9% in serious injuries, and only 0.7% in fatal injuries. This suggests that while most traffic accidents are not fatal, a significant portion still leads to injuries. Understanding which factors increase the likelihood of more severe outcomes—such as lack of seatbelt use or vehicle type—is central to our research.

The Seat Position Distribution chart shows that 76.9% of individuals were drivers, 12.4% were front-seat passengers, 8.7% were rear-seat passengers, and only a small fraction were motorcycle passengers.

Since injury risk can vary greatly depending on seat position (e.g., rear passengers may be less likely to wear seatbelts), this distribution helps us examine how position in the vehicle correlates with injury outcomes.

Lastly, the Vehicle Type Distribution shows that the majority of individuals were in cars (72.0%), followed by light commercial vehicles (11.7%), with other types like SUVs, motorcycles, and heavy vehicles making up smaller portions. Since different vehicle types offer different levels of protection in a crash, this variable is also closely tied to our research question on injury severity.

Overall, these charts provide foundational insights that support the next steps in our analysis, including exploring interactions among these factors and their combined effect on injury outcomes using regression and clustering models.



File 3. Proportion of Severity by Seat position (Grouped Bar Chart)

This grouped bar chart illustrates the relationship between seat position and injury severity level. For both drivers and front-seat passengers, the red bars representing non-injury account for the largest proportion, while the blue bars indicating fatal injuries are extremely small, nearly zero. This suggests that these positions are relatively safer, likely due to safety features such as seatbelts and airbags, which provide better protection in the event of a crash.

In contrast, for motorcycle passengers, the situation is very different. The proportions of serious injury (orange) and other injury (green) are significantly higher—around 40% and 45% respectively—while the non-injury rate drops sharply to just around 15%. This highlights a much higher risk of injury for motorcyclists, which is expected given the lack of enclosed protection in motorcycles.

Among all vehicle occupants, rear-seat passengers have the highest non-injury rate, even higher than drivers and front-seat passengers. This may suggest that, while often overlooked, the rear seat can be a relatively safer position—assuming seatbelts are used.

This chart is closely related to our research question, which explores how seating position affects the severity of injuries in traffic accidents. The visual evidence here supports the idea that seat position is a critical factor in injury outcomes. It also points to the importance of promoting safety measures for motorcycle passengers and encouraging seatbelt use in all seating positions, including the rear.

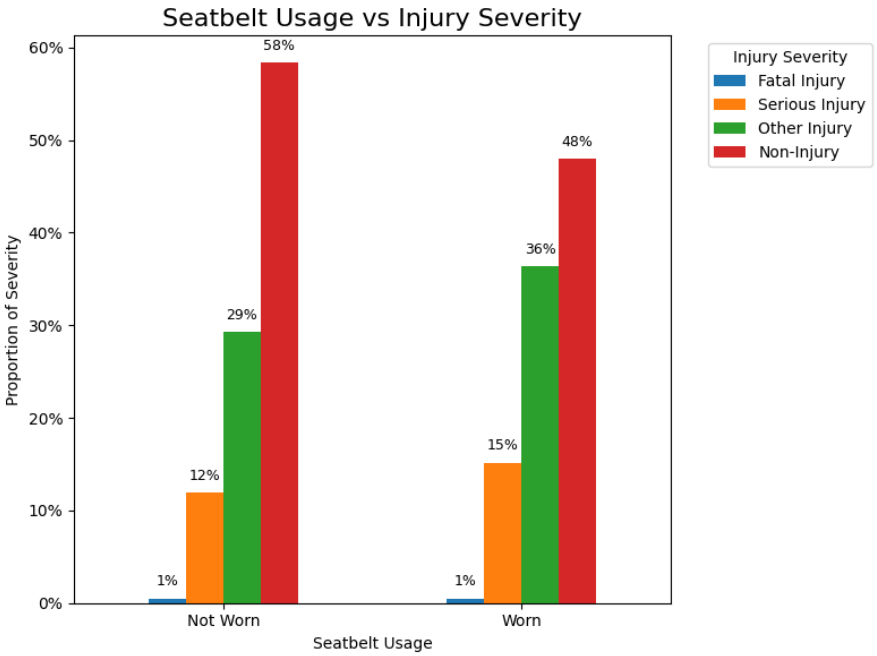


Figure 4 Proportion of Injury Severity by Seatbelt Usage (Grouped Bar Chart)

Figure 4 shows the relationship between use of seat belts and severity of injuries in traffic accidents. It can be seen from the bar chart that among the people who did not wearing seat belts, although 58% were not injured, 29% suffered other minor injuries, 12% suffered serious injuries, and 1% suffered fatal injuries. In contrast, among those wearing seat belts, the proportion of uninjured individuals dropped to 48%, but the proportions of minor injuries and serious injuries rose to 36% and 15% respectively, and the proportion of fatal injuries remained at 1%.

Although on the surface, the injury rate of the group wearing seat belts seems slightly higher, further verification through the pie charts in Figures 4.1 (seatbelts usage distribution and injury severity distribution) reveals that wearing seat belts effectively reduces the risk of fatal and serious injuries.

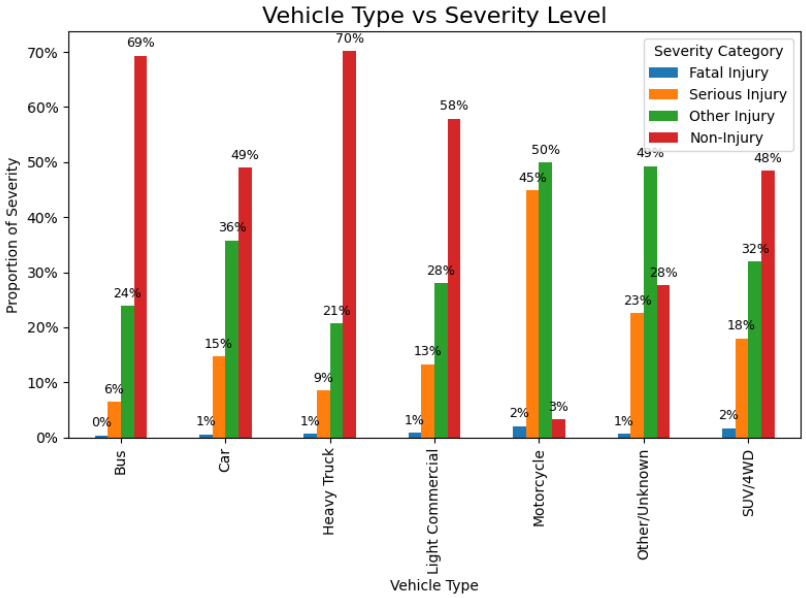


Figure 5 Proportion of Severity by Vehicle Type (Grouped Bar Chart)

Figure 5 shows the distribution of the severity of injuries in different vehicle types, including four categories: fatal injuries, serious injuries, other injuries and on injuries. It show significant difference in the distribution of injury severity among different vehicle type. This proportion of non-injury buses

and heavy trucks is the highest, reaching 69% and 70% respectively, indicating that the probability of these vehicles causing injures in traffic accidents is relatively low. However, the situation of motorcycles is particularly serious. The proportion of serious injuries is 45%, and the proportion of other injuries reaches 50%, which is much higher than that of other types of vehicles.

4.2 Supervised Learning

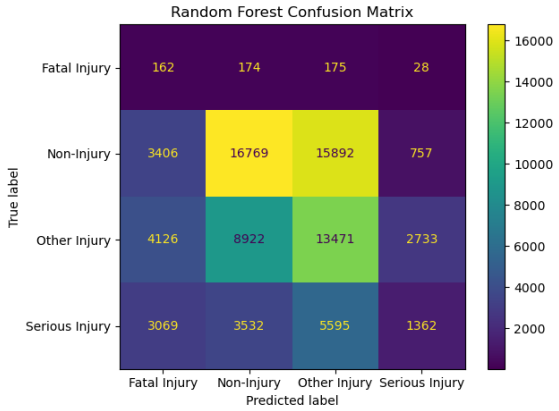


Figure 6: Random Forest Confusion Matrix

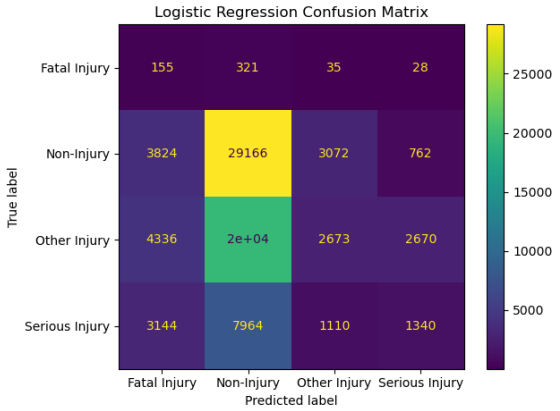


Figure 7: Logistic Regression Confusion Matrix

The analysis focuses on evaluating and comparing the performance of two supervised learning models—Logistic Regression and Random Forest classifier in predicting accident severity based on seatbelt usage, seating position and vehicle type.

Table 3: Classification report (Logistic Regression)

	precision	recall	f1-score	support
Fatal Injury	0.01	0.29	0.03	539
Non-Injury	0.51	0.79	0.62	36824
0ther Injury	0.39	0.09	0.15	29252
Serious Injury	0.28	0.10	0.15	13558
accuracy			0.42	80173
macro avg	0.30	0.32	0.24	80173
weighted avg	0.42	0.42	0.35	80173

Table 4: Classification report (Random Forest)

	precision	recall	f1-score	support
Fatal Injury	0.02	0.30	0.03	539
Non-Injury	0.57	0.46	0.51	36824
Other Injury	0.38	0.46	0.421	29252
Serious Injury	0.28	0.10	0.15	13558
accuracy			0.40	80173
macro avg	0.31	0.33	0.28	80173
weighted avg	0.45	0.40	0.41	80173

Both models show limited predictive performance, with overall accuracies of 42% for Logistic Regression and 40% for Random Forest. Given the imbalanced class distribution (e.g. only 0.67% of cases fall into the "fatal injury" category), these accuracy values should be interpreted cautiously.

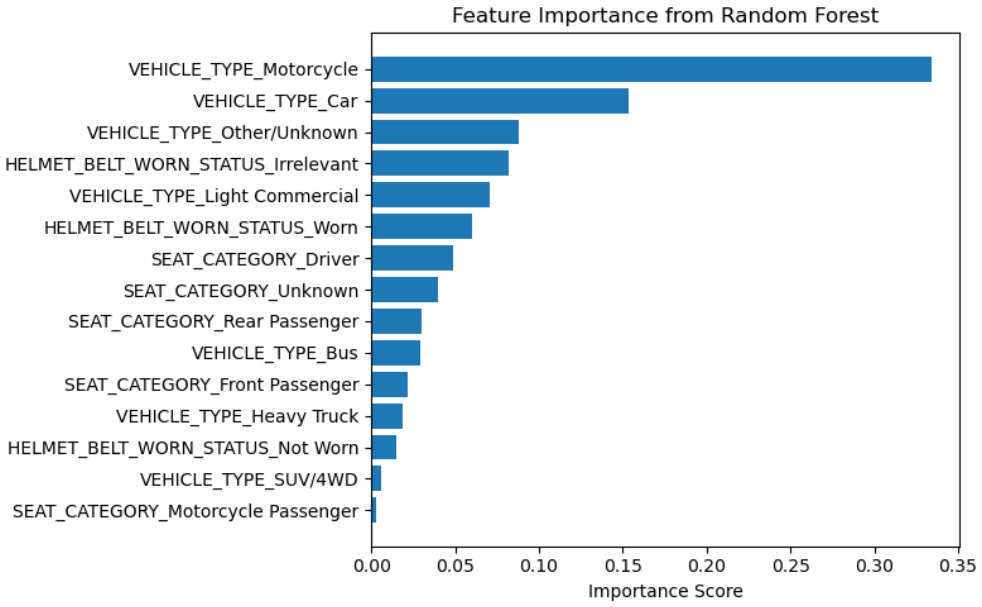


Figure 8: Feature Importance from Random Forest

The Logistic Regression achieved relatively better recall on the majority class (non-injury, recall = 0.79), but its performance for minority classes like "fatal injury" and "serious injury" remained poor (F1-scores of 0.03 and 0.15, respectively). This suggests the model is biased toward the majority class and lacks complexity to handle class imbalance effectively.

The Random Forest model demonstrated slightly better performance on "other injury" and "fatal injury" categories (recall = 0.46 and 0.30 respectively), suggesting it captures non-linear patterns better than logistic regression. However, its overall accuracy is slightly lower, indicating potential overfitting or confusion across similar classes.

The confusion matrices reveal common misclassifications:

- Many "other injury" and "serious injury" cases are misclassified as *Non-Injury*.
- A large portion of "fatal injury" cases are correctly identified in Random Forest (30%), but both models struggle with precision due to the small number of fatal samples.

These results indicate imbalanced classes and limited features may be key factors limiting performance.

Random Forest's feature importance analysis shows that seatbelt usage and seating position contribute more significantly to injury severity prediction than vehicle type. However, since these features alone are insufficient to distinguish between multiple injury levels, integrating additional variables—such as speed, weather, impact direction, or collision type may improve model discriminability.

4.3 Clustering

The application of clustering analysis resulted in the identification of three distinct behavioural profiles among road users, based on seatbelt usage, seating position, and vehicle type. The following analysis demonstrates the sample composition, injury outcome patterns, and feature distributions of these clusters, which serve as a basis for further interpretation.

The three clusters differed significantly in both sample size and injury outcomes. As demonstrated in Table 1, Cluster 0 exhibited a dominance over the dataset, comprising 367417 individuals (91.66%), followed by Cluster 2 with 29516 records (7.36%) and Cluster 1 with only 3930 records (0.98%), as shown in Table 1.

Table1: Sample Count and Proportion by Cluster

Cluster	Number of Samples	Percentage
0	367417	91.66%
1	3930	0.98%
2	29516	7.36%

Despite their relatively small size compared to cluster 0, clusters 1 and 2 exhibited disproportionately injury severity outcomes. The stacked bar chart in Figure 7 illustrates the distribution of injury severity across the three clusters, highlighting difference in risk exposure among groups.

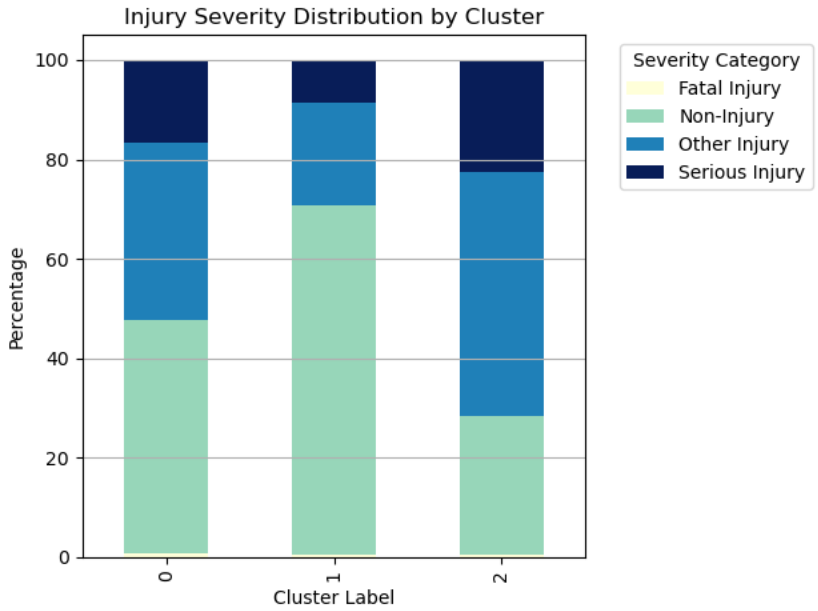


Figure 7. Injury Severity Distribution by Cluster (Stacked Bar Chart)

The following table (Table 2) presents the precise proportions of injury severity. Cluster 0 demonstrated approximately half of proportion of non-injury cases (47.12%), and comparatively low levels of serious injury (16.55%). Cluster 1, despite its modest size, exhibited the highest proportion of non-injury cases (70.20%), yet it demonstrated a lower incidence of serious and other injuries. Cluster 2 recorded a 27.91% occurrence of non-injury outcomes, with the majority of subjects experiencing other injuries (49.02%) and serious injuries (22.46%). That means it has the lowest proportion of "Non-Injury" but highest of "Serious Injury" and "Other Injury". Although the distribution of fatal injury is tiny for all 3 clusters, under 1%, Cluster 0 has the highest 0.68%.

Table 2. Injury Severity Distribution (%) Across Clusters

Cluster	Fatal Injury	Serious Injury	Other Injury	Non-Injury
0	0.68%	16.55%	35.65%	47.12%
1	0.59%	8.55%	20.66%	70.20%
2	0.60%	22.46%	49.02%	27.91%

With a view to interpreting the behavioural structure of each cluster, an analysis was conducted of the composition of seatbelt usage, seating position, and vehicle type. The distribution of features is shown in the cluster profile heatmap (Figure 8). The analysis about Injury Severity distribution is presented above and will not be repeated. Cluster 0 consists primarily of drivers (76.12%) operating cars (78.58%), with a substantial portion (60.67%) recorded as wearing seatbelts. However, more than 1/3 of the data has been allocated to "Irrelevant" for seatbelt usage (38.3%), showing considerable amount. Cluster 1 is composed exclusively of heavy truck occupants (100%), predominantly drivers (89.10%). Seatbelt

usage is mostly recorded as “Worn” (63.46%), followed by “Irrelevant” (34.20%). Approximately all of Cluster 2 falls into "Other/Unknown" vehicle types of category (99.14%), with considerable proportion of seatbelt usage recorded as “Irrelevant” (83.04%).



Figure 8. Cluster Profiles Heatmap: Seatbelt Status, Seat Position, Vehicle Type, and Injury Severity (% by Cluster)

This analysis only outlines the distributional characteristics of each cluster without attributing causality or interpreting risk levels, which would be addressed in Section 5.

5 Discussion and Interpretation

5.1 finding1： rear seat positions are relatively safer than front positions (in cars)

Frome the grouped bar chart on seat position verse injury, it can be observed that rear passengers have the highest proportion of non-injury outcomes, it is greater than drivers and front-seat passengers. Meanwhile, front passengers and drivers share similar injury severity distributions, with slightly higher rates of serious or other injuries than rear passengers.

The cause of this finding may be related to several safety factors as follows:

Frontal collisions are one of the most common types of crashes. Reat-seat passengers are a little farther from the impact zone of the accident than front-seat passengers. Therefore, in frontal collisions, rear-seat passengers have a higher probability of not being injured because they are farther from the point of impact.

In this report, we did not consider whether have airbags or not. As a result, the front seat passenger is more likely be injured by crushing dashboard, windshield and steering wheel than rear seat passengers. This finding is highly tied to our research question, which investigates how seat position relates to injury severity. It provides empirical support for promoting rear sear occupancy, and explained why children and elder people are suggested to seat in the rear seats, and

5.1 Finding 2: Seatbelt usage greatly improves safety outcomes

It can be seen from the grouped bar chart of seat belts usage and injuries level, although the proportion of injured people who are not wearing seat belts is relatively heigh, this phenomenon may be affected by factors such as sample size, seat position and vehicle type.

The reason for this discovery can be explained by the following factors:

1. Sample size: The sample size of people who did not wear seatbelts is relatively small.When the population is low, even a small amount of damage can lead to a higher injury percentage.
2. Seating position: Most people wearing seat belts are likely divers or front-row passengers, and they

were at a higher risk when the car crash occurred. Although they wore seat belts, they were close to common collision types, especially in the frontal collision, resulting in a higher injury rate.

3. Vehicle type: the type of vehicle plays an important role in the severity of injuries. Passengers in large vehicles benefit from stronger structural protection and higher seat positions, reducing the risk of injury. In contrast, smaller vehicles offer less protection during collisions, resulting in a higher injury rate.

From the perspective of government policy, this discovery highlights the urgent need to enhance public education activities on the use of seat belts. Traffic safety education programs should be widely promoted to raise people's awareness, especially for high-risk groups such as drivers and front-row passengers. Governments should emphasize the importance of wearing seat belts in all seats, ensure that individuals take proactive measures to protect themselves, and minimize preventable risks as much as possible.

Furthermore, the impact of vehicle types on the severity of injuries indicates that the government should actively promote the use of safer vehicles equipped with advanced safety features. Policy measures may include encouraging manufacturers to develop and produce safer vehicles, as well as introducing stricter vehicle safety assessment standards to ensure that the cars sold on the market meet high safety standards.

In sum up, the protective role of seat belts still cannot be ignored, especially in high-risk position and serious accidents, where seat belts can significantly reduce the degree of injury, Therefore, wearing a seat belt is one of the most effective ways to minimize the degree of injury.

5.1 finding 3: Vehicle type strongly influences injury severity

According to the grouped bar chart of "vehicle type verse severity level" which presents clear difference in injury patterns by vehicle type. The most "dangerous" vehicle is motorcycles, since it shows the highest rates of fatal injury (2%) and the lowest rates of non-injury (15%).

Cars and light commercial vehicles have moderate injury distributions, with nearly half of the individuals unharmed.

Buses and heavy trucks showed highest non-injury percentages (69% & 70%), and also showed the lowest proportion of serious injury rate (6% & 9%).

The cause of these differences may be due to the structural and physical factors as follows:

Motorcycles lack the kind of outer shell that four-wheel vehicles have, and because of their higher speed, braking in an emergency cannot stop them quickly. Therefore, in the event of a collision, due to the excessive speed and the lack of protective measures, the probability of injury is higher, and the severity of the injury is greater.

Although, cars provide outer shell and moderate protection, but their lightweight structure means occupants are still exposed to significant force.

Buses and trucks are larger and more robust, and their occupants sit higher above ground, often isolated from the direct point of impact. However, the bulky vehicle body may make it difficult for the entire car to turn, and the braking will be slower, thus imposing higher technical requirements on the driver.

This finding is also highly related to our research topic, supported the vehicle type affected the severity injury level. It underscores the importance of tailoring safety interventions to vehicle type. For instance, the motorcyclists must wear helmet, drive safer and slower. Car drivers benefit from continued improvements of airbag design. Heavy vehicles such as busses and heavy trucks drivers, it is essential to receive formal driving training. This explains why the requirements for this type of driver's license are more stringent.

5.2 Key Findings and Interpretations in model performance

- **Model performance reflects class imbalance**

The models' performance clearly reflects the underlying class imbalance, with predictive accuracy heavily skewed toward the majority non-injury class at the expense of minority injury categories.

Both Logistic Regression and Random Forest demonstrate this bias through their disproportionately high recall for non-injury cases (79% and 46% respectively) compared to 30% recall rates for serious and fatal injuries. This pattern emerges because standard machine learning algorithms optimize for overall accuracy in imbalanced datasets by preferentially classifying cases into the dominant category. The models' tendency to default to non-injury predictions reveal their fundamental inability to properly weigh the importance of rare but critical injury cases during training. While the Random Forest's marginally better performance on minority classes suggests some inherent advantage in handling imbalanced data through its ensemble structure, neither model achieves clinically or operationally useful performance for severe injury prediction. This underscores the necessity of implementing specialized techniques like other alternative evaluation metrics that don't disproportionately reward majority class accuracy before these models could be deployed in real-world safety applications where missing severe injuries carries substantial consequences.

- **Unexpected finding: Random Forest performed worse than Logistic Regression in overall accuracy**

Contrary to expectations, the Random Forest model underperformed Logistic Regression in overall accuracy (0.40 vs. 0.42), an unexpected result given Random Forest's typical superiority in handling complex datasets. This anomaly likely stems from the extreme class imbalance, where the ensemble method's bootstrapping approach may have inadequately represented minority classes in many decision trees, while Logistic Regression's simpler linear decision boundary proved more robust to the skewed distribution. The Random Forest has better recall for fatal injuries (30% vs 29%) but significantly worse precision (2% vs 1%), suggesting it overfit to rare cases while losing predictive power on the dominant non-injury class. These findings challenge model selection for imbalanced data and highlights how simpler algorithms may sometimes outperform more complex ones when critical features exhibit predominantly linear relationships with the target variable.

- **Misclassifications are dominated by “non-injury” predictions**

The models' systematic tendency to default to “non-injury” predictions highlights a critical failure mode where true injury cases are overwhelmingly misclassified as non-events, particularly for serious and fatal injuries where recall rates below 30% demonstrate dangerous severity underestimation. This pattern stems primarily from the combined effects of class imbalance (with non-injury cases comprising 45.9% of samples) and lacking feature representation of injury severity markers, causing the algorithms to optimize accuracy by favoring false negatives over false positives. While the Random Forest shows marginally better balance (46% recall for both non-injury and other injury cases versus Logistic Regression's 79%-9% split), both models exhibit unacceptable risk profiles for safety applications where missing severe injuries carries far greater consequences than unnecessary alerts. Addressing this bias requires not just technical adjustments but fundamental improvements in feature engineering to capture clearer severity signals and potentially alternative modeling approaches that prioritize recall on critical injury classes over aggregate accuracy.

5.3 Cluster-Based Behavioural Profile and Understanding

The clustering analysis revealed three distinct profiles of road users, each reflecting a particular combination of behavioural and structural characteristics. These clusters demonstrate that injury risk arises form interactions among vehicle type, seatbelt usage and seating position, rather than from any single factor. Instead of considering these features in isolation, the clustering process revealed configurations of behaviour that define meaning safety patterns.

Cluster 0, the largest cluster, reflects a profile of typical private road users. The individuals in this group predominantly engaged in driver roles and exhibited relatively consistent seatbelt usage. This combination appears to reflect standard safety behaviours in regulated environments. Regardless not being impervious to injury, this cluster exhibits a behavioural pattern that is associated with a comparatively reduced risk.9,

Cluster 1 is exclusively composed of heavy truck drivers, thus representing a group that is both professional and occupationally homogeneous. The utilisation of seatbelts was comparatively high among the three groups, resulting the highest proportion of “Non-injury” cases. However, under the surface of safety, heavy truck drivers are still considered to be high-risk road users due to their inherent characteristics. Thes include the operation of large vehicles, prolonged travel times, and the potential for fatigue or environmental exposure. The cluster of evident indicates that risk cannot be disregarded based on a single protective behaviour; rather, it must be understood within the operational realities of the situation.

Cluster 2 is indicative of a substantial and structurally vulnerable segment of the road user population. The present group is mostly composed of drivers in vehicles classified as “Other/Unknown”, with seatbelt usage categorised as “irrelevant”. It is important to consider which road users, such as e-scooter riders and cyclists, for example, may not be subject to the same regulations as other road users and therefore may not be required to wear seatbelts. In contrast to the other clusters, which demonstrate a greater alignment with conventional road safety patterns, this profile reflects a combination of ambiguous vehicle classification, low protection measures, and atypical road user roles. The consequences of these characteristics are evident in the markedly worse injury outcomes observed. It is crucial to not that the visibility of this group in the clustering results highlights that such high-risk configuration are not isolated anomalies but rather are integrated within a meaningful proportion of the dataset. The presence of these cases demands attention not only as statistical outliers, but also as indicators of systematic gaps in regulation, enforcement, and road safety education.

The significance of these observations lies in their capacity to illuminate the manner in which risk is structured within the population. This structuring is not merely a consequence of individual behaviours but rather is shaped by behavioural combinations that are themselves influenced by contextual factors. Clustering facilitates the identification of these structural patterns and the discovery of groups whose risk would otherwise remain concealed in aggregate analysis. This approach represents a departure from conventional one-variable-at-a-time explanation, thereby facilitating a more integrated understanding of road injury risk. Consequently, the behavioural profiles derived from clustering can directly inform policy by identifying which groups require targeted intervention, which conditions signal compounded risk, and how resources might be allocated more effectively to reduce severe injury outcomes across the road network.

6 Limitation and improvement

In order to concentrate the investigation on Seatbelt Usage, the HELMET_BELT_WORN column from the dataset is employed. Although the dataset provides information regarding the utilization of a helmet and the deployment of an airbag, these data were deemed to be “irrelevant” and thus excluded form further analysis. While this simplification enabled uniform encoding, it inevitably resulted in the loss of potentially meaningful distinctions between protective mechanisms. For instance, the use of helmets by motorcyclist or the implementation of airbag systems in vehicles may have a significant impact on the severity of injuries, factors which were not addressed in the present investigation. It may therefore be beneficial to reintroduce these features separately in future studies.

Another significant limitation exists in the combination of imbalanced cluster sizes and the assumptions underlying the KMeans algorithm. In the present analysis, the vast majority of samples were found to fall into Cluster 0, with Cluster 1 and 2 comprising much smaller portions of the dataset. The K-Means algorithm assumes spherical clusters separable using Euclidean distance, which may be problematic here as features we used are categorical and one-hot encoded. This transformation has the potential to artificially inflate the distance between semantically similar categories, thus leading to the conclusion that the Euclidean distance is a poor measure of true behavioural similarity. The algorithm may consequently exhibit overfitting of dominant groups, while compressing minority groups clusters or misallocating boundary cases. This structural imbalance, when combined with the geometric constraints of KMeans introduces a risk that the resulting clusters reflect artefacts rather than meaningful behavioural patterns. It is recommended that future research endeavours focus on the exploration of alternative clustering techniques that demonstrate a higher degree of compatibility with categorical or mixed-type data. Such techniques may include density-based or model-based algorithms. One-hot

encoding treats all categorical values as independent features, but certain categories may interact (e.g. front seat and no belt might be more dangerous than rear seat and no belt). Our models did not explicitly capture such interactions.

7 Conclusions

This report explored the relationship between injury severity in traffic accidents and key factors such as seatbelt usage, seat position, and vehicle type. And the relationship between these key factors and severity injury level. And identify high-risk road users through clustering. By combining multiple datasets and applying classification, visualization, and clustering techniques, we identified several important patterns. In this study, by combining the two datasets of person and vehicle, key variables (such as seat belt usage, seat position, vehicle type and injury degree) were selected, missing value processing and feature classification were carried out, and One-Hot encoding and standardization were used for data preparation. The analysis methods include chart visualization, mutual information heat map, chi-square independence test, K-Means clustering (the number of clusters is determined using the Elbow and Silhouette methods), etc. This process ensures that the research conclusion is supported by data and has explanatory power. In data analysis stage, we mainly used grouped bar charts, pie charts, heat maps, and elbow k methods. Grouped bar charts can enable comparisons among multiple types of data, while the data of each factor is also very clear, and the trends of different data can also be seen. Pie charts show the proportion of different factors in a better way. A heatmap used in our report visually represents the correlation between variables using color intensity (such as through Normalized Mutual Information, NMI), making it easy to identify which variables share the most information. In this research, since seatbelt usage, seat position, vehicle type, and injury severity are all categorical variables, the heatmap helps us understand the strength of interaction between these factors. Through these data and methods support, we mainly come up with various interesting findings, including rear seat positions are relatively safer than front positions (in cars), seatbelt usage greatly improves safety outcomes, cluster-Based Behavioral Profiles that identified three distinct road user groups, each characterized by unique combinations of vehicle type, seatbelt usage, and seating position. For models, Model performance reflects class imbalance, Random Forest performed worse than Logistic Regression in overall accuracy, Misclassifications are dominated by “non-injury” predictions. These findings are all discussed through causes with data analysis and significance of finding it. Finally, we provide our suggestions through our findings. Including, strengthening education on traffic safety, making all passengers fasten their seat belts, and passengers of different groups should sit in different positions in the vehicle, for instance, children and elder people should sit in the back. Drivers and passengers of different vehicle models have different requirements. For decision-makers, we should ensure the strictness and rigor of obtaining driver's licenses.

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